and supplies needed fuel. The petrol pressure exerted from the mechanical type is directly proportional to the spring in the diaphragm.

Usually in the mechanical petrol pump the pressure in the petrol depends on the tension in the spring of the diaphragm. Generally the pressure of the petrol coming out from the petrol will be in the range of 1 kg/cm^2. Fig 9.3.1.1 shows the diagram of A.C. Mechanical Fuel Pump.

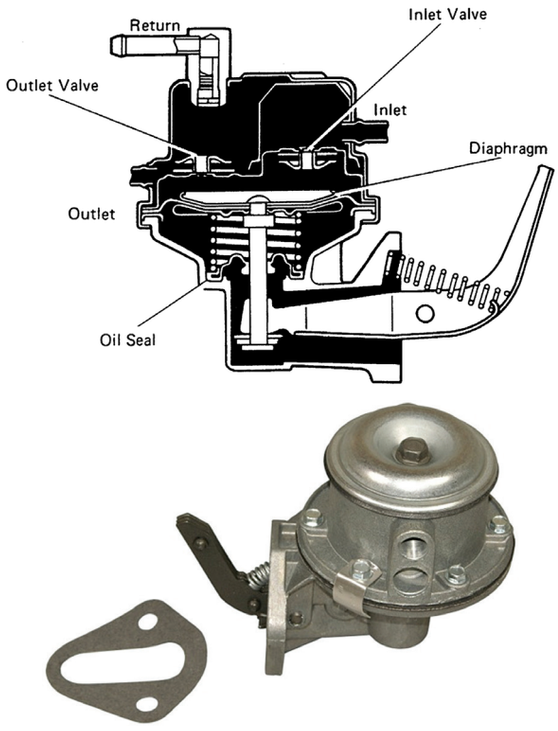


Figure 9.3.1.1 AC Mechanical Petrol Pump

9.3.1.2 SU Electrical fuel pump

As like the AC mechanical petrol pump SU electrical petrol pump also will be actuated by the diaphragm action. But in this pump, electricity is used to move the diaphragm up and down instead of eccentric. When the driver switch ON the ignition, the electricity is passed from the battery to the solenoid winding due to that the solenoid windings get demagnetised. Due to which the armature in the diaphragm moves upward. So that Vacuum is being created in pump chamber, by that time the suction valve gets opened and petrol sucked inside. At this stage, the armature moves upwards and the breaker point moves away and the electricity is disconnected. Then, by the spring action the diaphragm moves to the original position due to which the petrol in the pump chamber comes out by outlet valve and reaches float chamber in carburettor. If the fuel in the float chamber is full then the needle valve closes the inlet valve due to this the high pressure in pump chamber the spring restrict the action of diaphragm. When the level of fuel in the float chamber is reduced the needle valve gets opened and fuel (petrol) from the pump gets in as before. Fig 9.3.1.2 shows the diagram of S.U. Electrical Fuel Pump.



Figure 9.3.1.2 SU Electrical Fuel Pump

9.4 AIR FUEL RATIO (AFR)

For the various load and speed of the engine, the air and fuel should be mixed and supplied in the required manner with

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